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Disassortative dynamic BA models inspired by the Bitcoin Lightning Network

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***Abstract*—The evolution of the *channel graph* of the Lightning Network, the main layer-2 solution on top of Bitcoin, has been often modeled as a Barabási-Albert random graph. However, the available data on the channel graph of the Lightning Network indicate that it is stabilizing over a network structure with negative *assortativity*, while the assortativity of a BA random graph tends to zero as the number of nodes grows to infinity.**

In the BA model, new edges are created when a new node joins the network and they never disappear. In the channel graph of the Lightning Network, channels can be *closed* by one or both of its endpoints at any time. In this paper, we propose two dynamic versions of the BA-model in which edges can disappear at any time. In the first version, the edge disappearance rate is a function of the degrees of the endpoints, in the second one it is a function of the edge *capacity*. Simulations show that, in both models, the assortativity converges to a negative value, that depends on the edge disappearence rate. Our results suggest that the disassortative nature of the Lightning Network, as well as that of several other real networks, could be a consequence of the dynamic nature of the edges of the network.

***Index Terms*—Dynamic random graphs, Barabási-Albert model, Assortativity, Bitcoin Lightning Network, Simulations**

I. Introduction

The Bitcoin protocol, as designed by Satoshi Nakamoto [1] and coded in the current reference implementation [2], imposes a strict limit on the size of each block of the Blockchain. On the one hand, such a limit makes the Blockchain grow at a sustainable rate, so that it can be easily stored by tens of thousands of nodes (see, e.g., [3] for an updated estimation of their number), the so called *full-nodes*, that have the entire Blockchain and can thus independently evaluate the validity of all transactions and blocks. On the other hand, the bound on the blocks' size directly implies that only a limited number (on average less than ten) of transactions can be confirmed (i.e., included in the Blockchain) every second.

The Lightning Network [4] is a layer-2 protocol designed on top of Bitcoin that allows bitcoin owners to use them as a reserve value [5], forming a network of *payment channels* in which payments among agents can be executed in a safe and scalable way. Namely, two parties can lock an amount of their bitcoins into a 2-of-2 multisignature address in a way that (i) the balance of bitcoins owned by the two parties can be updated by exchanging private messages between them and (ii) each party at each time has a transaction signed by the other party that, if broadcasted to the Bitcoin network, spends the funds locked in the 2-of-2 multisig assigning each party her current share of the balance and recording it on the Blockchain. The operation of locking funds in the 2-of-2 multisig address is referred to as *opening a channel*; the operation of broadcasting the transaction that records on the Blockchain the current balance is referred to as *closing the channel*. The overall amount of funds locked in the 2-of-2 multisig address is the channel *capacity*; a *payment* is an update of the balance. The Lightning Network protocol specifies the cryptographic mechanisms that make it possible to update a channel balance in a safe and trustless way for both parties and the mechanisms that allow a payment to be routed accross channels. We refer the reader interested in the details of such mechanisms to [6].

In this paper we are interested in the design and analysis of random graph models inspired by the network-formation process of the channel graph of the Lightning Network. The evolution of the network of channels in the Lightning protocol is driven by several different forces: a user that creates a new channel in order to be able to execute payments typically would like to minimize the *fees* she incurs in, hence she will likely choose to open a channel with one or more highly *central* nodes; a merchant that opens one or more channels to receive payments typically wants to be sure to have sufficient *inbound liquidity* in order to allow all its potential customers to easily find short and cheap payment paths; a routing node typically decides how to open channels in a way that maximizes its expected incoming fees. It has been observed that these driving forces tend to shape the network as a scale-free graph [7], [8]. Understanding and predicting how the network of channel is going to evolve in time is a major challenge and can be useful to design more efficient path-finding algorithms for routing payments [6].

Power Law

The classical Barabási-Albert model [9] has been used as a mathematical model of the Lightning Network [10], [11]. In that model, at every discrete round a new node is added to the network and such a node selects one or more neighbors to connect to, each node chosen independently at random according to a probability distribution that gives more weights to nodes with higher degree. The degree distribution of the Barabási-Albert model is consistent with the degree

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distribution observed in the Lightning Network [12]. However, in [8], [12] the authors noted that the *assortativity* coefficient of the Lightning Network seems to converge to a negative value, while the assortativity coefficient of the BA-model tends to zero as the number of nodes grows to infinity [13]. The assortativity coefficient, first introduced in [13], serves as a measure of the tendency of nodes within a network to form connections with other nodes that are similar according to a node property. Specifically, we focus on *degree assortativity*, which reflects the propensity of nodes to link with others of comparable degree. Throughout this paper, any mention of 'assortativity' implicitly refers to degree assortativity. The assortativity coefficient is defined as the Pearson correlation coefficient of the excess degree between pairs of connected nodes. Positive values of this coefficient suggest a network where nodes of similar degrees are more likely to interconnect, characterizing the network as *assortative*. Conversely, negative values indicate a propensity for nodes of differing degrees to link, rendering the network *disassortative*.

A. *Our contribution*

↳ LN lo ha a 0

One of the features that make the Lightning Network an useful testing workbench for studying dynamic network models is its *availability*: The payment channels in the Lightning Network are publicly announced to all nodes (that is needed to allow nodes to find routing paths for payments), hence the network of channels, as well as further information like, e.g., the fees applied for each channel, is known to any node participating in the network. We installed a Lightning Network node and started collecting all of the announcements to be able to reconstruct the snapshots of the network at any point in time. For the past evolution of the Lightning Network, we relied on the data made available in [14]. We used all the collected data to gain some insights for the design of appropriate random graph models (see Section III).

Motivated by the observations emerging from the evolution of the Lightning Network, in this paper we introduce two variants of the Barabási-Albert model in which edges can disappear randomly. In the first variant, the edge-disappearance rate depends on the degrees of the two endpoints (see Section IV). In the second variant, the edges are weighted and the edge-disappearance rate depends on the weights: more precisely, the probability that an edge disappears depends on the ratio between the weight of the edge and the sum of the weights of the edges incident on the endpoints (see Section V). In both models the edge-disappearance rate is tunable by means of one single parameter β.

By running extensive simulations, we show that our simple variants of the BA-model preserve the degree distribution and, moreover, they exhibit an assortativity coefficient that tends to a negative value, as the number of nodes grows. Such a value is a function of the tuning parameter β in the models. This result is remarkable since, while it is possible to design random graph models with a specified assortativity coefficient [15], such models are typically "centralized" and not compatible with real networks like the Lightning Network, in which the topology is not *designed* by a central authority but it *emerges* as a result of the individual choices of the nodes participating in the network and the criteria they use to open channels with other nodes. Our simulations show that it is sufficient to appropriately include an edge-disappearence rate in the BA-model to have a random graph that tends to have negative assortativity. Hence, our results suggest that the disassortative nature of the network might emerge as a consequence of edges disappearing during the evolution of the process.

B. *Related work*

The Lightning Network, initially proposed to mitigate the scalability issues of Bitcoin, has been a subject of extensive research and scrutiny, mostly focused on its underlying vulnerabilities and challenges. A significant concern pointed out in [7] is the network's tendency towards centralization, which could potentially compromise privacy by allowing certain entities to monitor transactions. Privacy and security within the Lightning Network have been the focus of several studies. For example, research has explored conducting transactions with enhanced privacy [16] and introduced a novel geometric framework for analysis [17]. Another study addressed the vulnerability of payment channels to disruption by improperly formatted transactions [18].

In an effort to understand the Lightning Network's dynamics, the development of a diagnostic tool, the "Time Machine", has enabled researchers to collect and analyze temporal data [14]. In [12], the authors compared the Lightning Network to the Barabási-Albert model of the same size over the period of the network's development. They observed that nodes tend to connect with nodes closer to higher centrality, more so than to nodes with higher degree centrality or betweenness centrality. They also noticed a significant gap in diversity and diameter between the Lightning Network and the Barabási-Albert model of the same size.

The dynamic network models we introduce in this paper are mostly inspired by the Bitcoin Lightning Network. However, several other real networks turn out to be disassortative and a finding satisfying explanation for such phenomenon has been the subject of extensive research [19], [20].

II. Preliminaries

Assortativity, or assortative mixing, is a concept first introduced in [13] to describe the tendency of nodes in a network to establish links between other nodes that are similar in some way. For example, in social networks, people tend to form relationships with peers of similar age. In contrast, in technological networks like the internet, most connections are between giant servers and smaller user devices. In the first case, we say that such a network is *assortative*, while in the latter, it is *disassortative*. The concept of assortativity is defined out of some kind of similarity between nodes. This similarity could be derived by using node features (if present) or from topological properties like node degree. In this paper when we use the term assortativity we implicitly refer to assortativity based on node degree.

Assortatività di una rete:

The assortativity of a graph is measured as a coefficient between -1 and 1 (it is the Pearson correlation coefficient [21], [22] of the degrees of the endpoints of a random edge of the graph). Formally, let $G = (V, E)$ be an undirected graph, then the assortativity coefficient of G is defined as

$$r = \frac{\sum_{jk} jk(e_{jk} - q_j q_k)}{\sigma_q^2}$$

Prob di avere un arco u.a.r. con un endpoint u.a.r. con grado k+1

where q_k is the probability that, when we pick an edge uniformly at random (u.a.r.) in G and one of its endpoints u.a.r., we end up in a node with degree $k+1$ (i.e., q is the *remaining degree* distribution of a random endpoint of a random edge in G), and e_{jk} and σ_q are the joint distribution and the standard deviation of the remaining degrees, respectively. Quantity e_{jk} is symmetric (for undirected graphs) and it satisfies $\sum_{jk} e_{jk} = 1$ and $\sum_j e_{jk} = q_k$.

III. Data Collection and Analysis

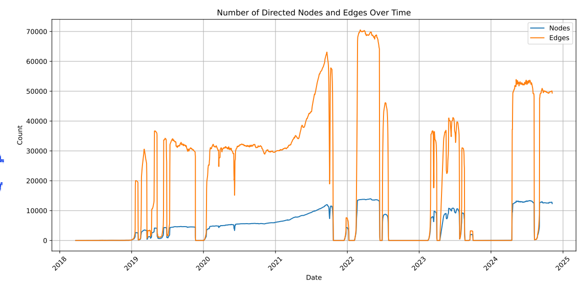
In this section, we detail the methodology employed to collect and analyze real-world data from the LN. Our analysis relies on two primary datasets: An *external* dataset, as provided in [14], that spans from March 20th, 2018, to September 23rd, 2023, with some gaps (see Figure 1a), and our *internal* dataset, that we are continuously building, since April 17th, 2024, using our own LN node.

A. Data collection tools

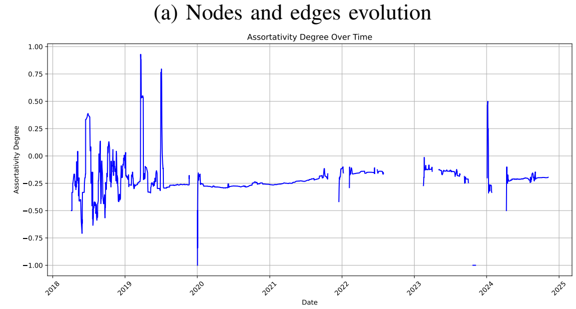
Every node participating in the LN receives information about every change in the network, e.g., channel openings, updates, and closures, fee updates, etc. In order to capture and analyze the LN's dynamics, we thus installed an LND (LN Daemon) client [23] and we developed some custom tools to perform the following key functions: (1) monitoring the LN for changes, recording all relevant information regarding the network topology and storing them in a way that on the one hand it is space efficient and on the other hand it allows us to easily retrieve relevant data; (2) reconstructing the network topology at any point in time from the stored data and analyzing the network dynamics and the evolution of key quantities like the degree assortativity.

B. Data analysis

An analysis of the LN revealed significant changes in its topology over the observed period. The number of nodes increased markedly, reaching its peak in March 2022, and was almost stable thereafter, ranging from almost 12,100 on April 17 to 12,345 on November 5, with some peaks up to 13,000 and occasional drops below approximately 12,000. However, despite the stable node count, the number of channels experienced a substantial decline—from approximately 70,000 during February to June 2022 to about 50,000 between April 17 and November 6, representing a reduction of about 28%. This indicates that while nodes continued to participate in the network, they maintained fewer connections on average (see Figure 1a). Additionally, the network's assortativity declined from -0.16 to -0.19, with the latter recorded on November



(a) Nodes and edges evolution

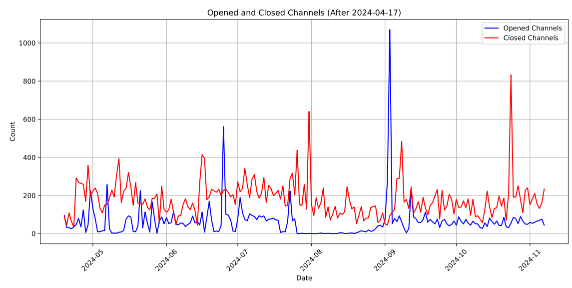


(b) Degree assortativity evolution

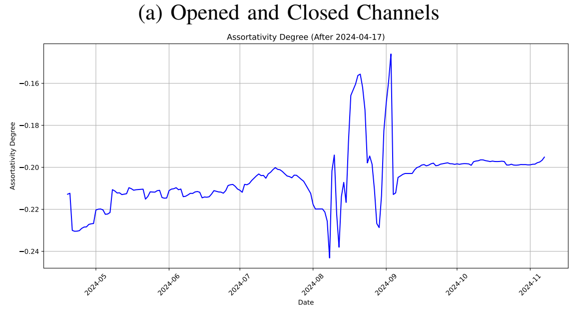
Fig. 1: Evolution of degree assortativity and number of nodes and edges in the LN over time

6, 2024 (see Figure 1b). The decreasing assortativity suggests an increasing disassortative mixing, meaning nodes are more likely to connect with others that are different in terms of connectivity degree. These trends imply a tendency toward a network structure characterized by highly connected hubs linked to many low-degree nodes, which could have implications for network efficiency, resilience, and decentralization.

Figure 2a and Figure 2b illustrate a key characteristic of the network: its dynamic nature. The network undergoes daily updates, with new channels being added and others closed, resulting in continuous shifts in assortativity over time. This phenomenon is depicted in Figure 2a, showcasing how the network structure evolves across the specified timeframe from April 2024 to November 2024 (data from August 8th to September 3rd are missing due to a technical issue that resulted in data loss for that time window). During this period, the average number of channels opened daily is approximately 82, while the average number of channels closed daily is around 135. This trend may be attributed to market fluctuations or strategic behaviors by certain nodes, such as closing and reopening channels to withdraw liquidity for reinvestment or to exit the network entirely. In contrast, Figure 2b highlights a subtle variation in assortativity, beginning at -0.172 on the first day and ending at -0.195 on the final day, with recorded minimum and maximum values of -0.193 and -0.153, respectively. This pattern emphasizes an heterogeneous structure in which high-degree nodes are more likely to connect with low-degree nodes. Such negative assortativity has an impact on both routing efficiency and network resilience. Moreover,



(a) Opened and Closed Channels



(b) Evolution of Degree Assortativity

Fig. 2: Evolution of degree assortativity and number of nodes and edges in the LN after 2024-04-17

it also confirms the tendency to increase centralization around core hubs (as observed some years ago in [8]), a phenomenon that can be attributed to the nature and behavior of the nodes, given that channels represent economic relationships between two nodes.

IV. A Barabási-Albert model with degree-dependent edge disappearance rate

The Barabási-Albert model [9] is one of the most well-known random graph models used to describe the growth of real networks in a simple way. This model is particularly successful because it exhibits a powerlaw degree distribution which is especially common in many networks. However, as noted in [13], the Barabási-Albert model has an assortativity coefficient that tends to 0 for n that goes to infinity. This goes in stark contrast with typical real networks which usually have non-zero assortativity. In Section III we observed that the Bitcoin Lightning Network has shown tendency towards negative assortativity during its evolution. (see Figures 1b and 2b) Specifically, it has been oscillating around a value of -0.2, and it seems it is slightly increasing in the last few months. In this section we define a variation of the Barabási-Albert model that still produce a network with powerlaw degree distribution while exhibiting a negative value of assortativity, which is also tunable by its parameters.

A. Model

The model has four parameters: n is the number of nodes, d is the number of neighbours each node connects to, β is the parameter that controls the probability for an edge to disappear, and the initial graph $G_0 = (V_0, E_0)$. At each subsequent round i the graph G_{i+1} is determined as follows:

- The $(i+1)$-th node is added to the network: $V_{i+1} = V_i \cup \{v_{i+1}\}$;
- The $(i+1)$-th node chooses d nodes to connect to, $u_1, u_2, ..., u_d$, each node u_j is chosen with probability $\frac{\delta_{G_i}(u_j)}{2|E_i|}$, where $\delta_{G_i}(u_j)$ denotes the degree of node u_j in the graph G_i: $E_{i+1} = E_i \cup \{\{v_{i+1}, u_1\}, \{v_{i+1}, u_2\}, ..., \{v_{i+1}, u_d\}\}$;

Più archi ha più viene scelto

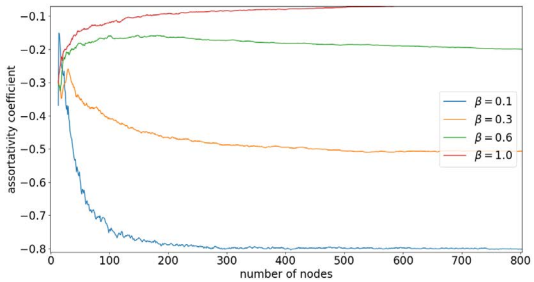
- Each edge $\{u, v\} \in E_i$ disappears with probability
$$q_i(u,v) = e^{-\beta \cdot max\{\delta_{G_i}(u), \delta_{G_i}(v)\}},$$
independently of the other edges and of the other rounds.

Our model is an extension of the Barabási-Albert model, allowing the possibility for edges to disappear. This possibility more accurately reflects the behavior of real-world networks, where connections often dissolve over time. The Barabási-Albert model is based on the principle of preferential attachment, where new nodes are more likely to connect to already well-connected nodes. The idea of using a degree-dependent edge disappearence probability is coherent with the principle of preferential attachment in the Barabási-Albert model, which implies that high-degree nodes are more "important" than low degree nodes and, therefore, less prone of losing connections.

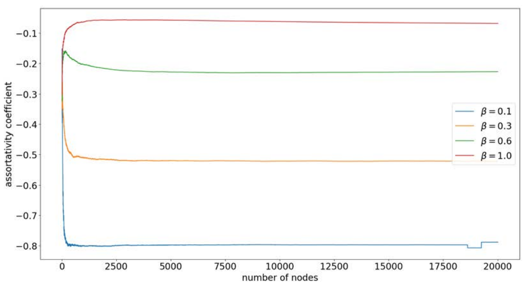
B. Simulations

We simulated the evolution of this model in various independent experiments and we recorded the results in term of assortativity, degree distribution, diameter, and number of edges, for different values of β. We used a complete graph with $d+1$ nodes as the initial configuration G_0.

The simulations were done using the C++ programming language and the g++ 12.2.0 compiler. The simulation takes in input the parameters d and β and the number of iterations of the model. At each round i, the choice of the d neighbors of node i is done efficiently by sampling a random node in the edge list representation of the graph. This effectively guarantees that each node v is selected with probability exactly $\frac{\delta_{G_i}(v)}{2|E_i|}$ (this idea was first introduced by Batagelj et al. [24] in 2005). Then, we computed and recorded to a csv file the value of n, m, assortativity and diameter of G_i for each i for that given simulation. We set the value of $d = 6$ and the number of iterations to $20,000$ and different values for β. For different β we executed 10 simulations and averaged the results across experiments with the same value of β. We observed that the assortativity converges to a negative constant value that is dependent on β. When β is higher than 1.0 the assortativity goes to 0 and the network more closely resembles a standard Barabási-Albert graph. However, as β goes to 0 the assortativity of the resulting network increasingly drifts towards a negative value. As we can see in Figure 3a, just by adding the edge deletion routine in the model we have that the assortativity converges steeply to a negative constant value dependent on the value of β. And as shown in Figure 3b, once it converges, the assortativity does not change significantly over subsequent iterations.



(a) Up to 800 nodes



(b) Up to 20000 nodes

Fig. 3: Evolution of degree assortativity of our model with different values of β

V. A weighted Barabási-Albert model with capacity-dependent edge-disappearance rate

In the model we presented in Section IV, the edges are not weighted and the disappearance rate of an edge depends on the degrees of the endpoints. In this section we define an edge-weighted version of the Barabási-Albert model in which edges have integer-valued *weights* (the *capacities*, in the context of the Lightning Network) and they disappear during the time-evolution of the process at a rate that depends on the weight. It turns out that, also in this edge-weighted model, the assortativity at which the dynamic graph tends to can be tuned by adjusting the rate at which edges disappear.

A. Model

The evolving random graph model $\mathcal{G}$ is defined by four parameters $\mathcal{G} = \mathcal{G}(G_0, h, d, \beta)$ with the following meaning. Starting from an initial graph G_0[1], at each round $t \in \mathbb{N}$ the following happens:

1) N_t new nodes join the network, where $\{N_t : t \in \mathbb{N}\}$ is a sequence of i.i.d. random variables with expectation h (in the simulations we consider geometric random variables);
2) Each new node u selects d nodes to connect to, with each node v chosen with probability proportional to the sum of the capacities of the edges incident on v;
3) For each new node u and for each chosen neighbor v, the weight of the edge $\{u, v\}$ is a random variable $C_{\{u,v\}}$ (in our simulations we set $C_{\{u,v\}}$ to be either binomial or power law);
4) Each edge $e = \{u, v\}$ disappears with probability
$$p(e) = \left[\frac{w(e)}{w(u) + w(v)}\right]^{\beta}$$
where $w(e)$ is the weight of edge e and $w(u)$ and $w(v)$ are the sums of the weights of the edges incident on nodes u and v, respectively.

Notice that, for $\beta = 0$, every edge disappears with probability 1 in one round, while for β that goes to infinity the model converges to the standard BA-model where edges never disappear.

B. Simulations

In order to compare our model with the evolution of the assortativity in the real Lightning Network (see Figures 1b and 2b), in this section we consider one *round* of the model corresponding to one *day* of the evolution of the network.

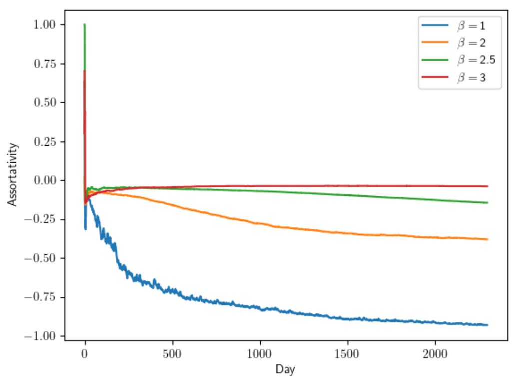
Figures 4a and 4b show the assortativity evolution for different values of β, where the other parameters of the model are set as follows: The initial graph G_0 was chosen as a complete graph with 10 nodes. The number of new nodes joining the network each day is a geometric random variable $G(p)$ with $p = 1/10$, so that the average number of new nodes per day is 10 and the total number of nodes and edges after about two thousand days matches the current number of nodes and edges in the Lightning Network. When a new edge is created, its weight is chosen according to a random variable (binomial in the case of Fig. 4a and power law in the case of Fig. 4b) with an expected value of ten million, so that the expected edge capacity in the model matches the current average capacity in *satoshi* of the Lightning Network.

Each plot in Figure 4 represents one single run of the model for about two thousand days. The plots clearly show that the trend of the assortativity is determined by the value of β, i.e., the parameter regulating the edge disappearance rate: The higher the value of β, the smaller the probability of an edge to disappear, the more the disassortative nature of the network diminishes. Even for $\beta = 3$, we can already observe a trend comparable with the one of the standard BA-model, while for $\beta = 1$ the network tends to be highly disassortative. The value of β that makes the network assortativity converge to a value close to the one observed in the real data, i.e., approximately -0.2, seems to be located somewhere between 2 and 3. The actual distribution of the edge capacities (binomial in Fig. 4a and power law in Fig. 4b) does not seem to have any qualitative impact in the evolution of the assortativity.

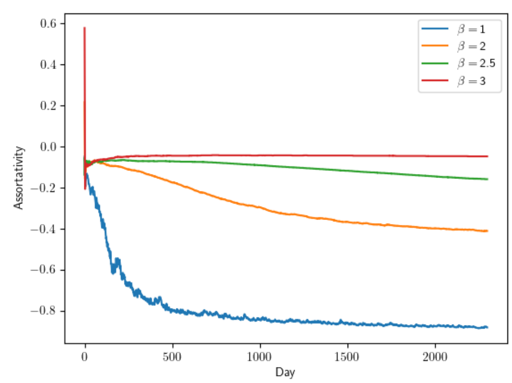
VI. Conclusions

In this paper we introduced two variants of the Barabási-Albert random graph model in which edges can also disappear. In the first variant the disappearance rate of an edge is a function of the degrees of its endpoints, in the second one

[1]In the simulations we take a complete graph with 10 nodes as G_0, with all edge capacities set to a constant $c \in \mathbb{N}$



(a) Edge capacities chosen according to a binomial probability distribution



(b) Edge capacities chosen according to a power law probability distribution

Fig. 4: Evolution of degree assortativity of our model with different values of β

it is a function of the capacity of the edge. In both cases, the edge disappearance rate is tuned by one single parameter and our simulations show that the assortativity is directly determined by that parameter. Our results thus suggest that the negative assortativity observed in the Lightning Network could be explained by the channel closures that periodically occur in that network.

References

[1] S. Nakamoto, "Bitcoin: A peer-to-peer electronic cash system," https://bitcoin.org/bitcoin.pdf, 2008.
[2] S. Nakamoto and et Al., "Bitcoin core," https://github.com/bitcoin/bitcoin, 2008, accessed: 2024-11-12.
[3] A. Yeow, "Global Bitcoin Nodes Distribution," https://bitnodes.io/, 2013, accessed: 2024-11-12.
[4] J. Poon and T. Dryja, "The bitcoin lightning network: Scalable off-chain instant payments," 2016, white paper. [Online]. Available: https://lightning.network/lightning-network-paper.pdf
[5] S. Ammous, *The bitcoin standard: the decentralized alternative to central banking*. John Wiley & Sons, 2018.
[6] A. Antonopoulos, O. Osuntokun, and R. Pickhardt, *Mastering the Lightning Network: A Second Layer Blockchain Protocol for Instant Bitcoin Payments*. O'Reilly, 2021. [Online]. Available: https://github.com/lnbook/lnbook
[7] P. Zabka, K.-T. Förster, C. Decker, and S. Schmid, "A centrality analysis of the lightning network," *Telecommunications Policy*, vol. 48, no. 2, p. 102696, 2024. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0308596123002070
[8] S. Martinazzi and A. Flori, "The evolving topology of the lightning network: Centralization, efficiency, robustness, synchronization, and anonymity," *Plos one*, vol. 15, no. 1, p. e0225966, 2020.
[9] A.-L. Barabási and R. Albert, "Emergence of scaling in random networks," *Science*, vol. 286, no. 5439, pp. 509–512, 1999.
[10] F. Engelmann, H. Kopp, F. Kargl, F. Glaser, and C. Weinhardt, "Towards an economic analysis of routing in payment channel networks," in *Proceedings of the 1st workshop on scalable and resilient infrastructures for distributed ledgers*, 2017, pp. 1–6.
[11] V. Davis and B. Harrison, "Learning a scalable algorithm for improving betweenness in the lightning network," in *2022 Fourth International Conference on Blockchain Computing and Applications (BCCA)*. IEEE, 2022, pp. 119–126.
[12] Z. Wang, R. Zhang, Y. Sun, H. Ding, and Q. Lv, "Can lightning network's autopilot function use ba model as the underlying network?" *Frontiers in Physics*, vol. 9, p. 794160, 2022.
[13] M. E. J. Newman, "Assortative mixing in networks," *Phys. Rev. Lett.*, vol. 89, p. 208701, Oct 2002. [Online]. Available: https://link.aps.org/doi/10.1103/PhysRevLett.89.208701
[14] C. Decker, "Lightning network research – topology datasets," https://github.com/lnresearch/topology, accessed: 2024-11-12.
[15] B. Zhou, X. Lu, and P. Holme, "Universal evolution patterns of degree assortativity in social networks," *Social Networks*, vol. 63, pp. 47–55, 2020.
[16] C. Pérez-Sola, A. Ranchal-Pedrosa, J. Herrera-Joancomartí, G. Navarro-Arribas, and J. Garcia-Alfaro, "Lockdown: Balance availability attack against lightning network channels," in *Financial Cryptography and Data Security: 24th International Conference, FC 2020, Kota Kinabalu, Malaysia, February 10–14, 2020 Revised Selected Papers 24*. Springer, 2020, pp. 245–263.
[17] A. Biryukov, G. Naumenko, and S. Tikhomirov, "Analysis and probing of parallel channels in the lightning network," in *Financial Cryptography and Data Security*, I. Eyal and J. Garay, Eds. Cham: Springer International Publishing, 2022, pp. 337–357.
[18] G. van Dam, R. A. Kadir, P. N. E. Nohuddin, and H. B. Zaman, "Improvements of the balance discovery attack on lightning network payment channels," Cryptology ePrint Archive, Paper 2019/1385, 2019, https://eprint.iacr.org/2019/1385. [Online]. Available: https://eprint.iacr.org/2019/1385
[19] S. Johnson, J. J. Torres, J. Marro, and M. A. Munoz, "Entropic origin of disassortativity in complex networks," *Physical review letters*, vol. 104, no. 10, p. 108702, 2010.
[20] A. Topirceanu, M. Udrescu, and R. Marculescu, "Weighted betweenness preferential attachment: A new mechanism explaining social network formation and evolution," *Scientific reports*, vol. 8, no. 1, p. 10871, 2018.
[21] K. Pearson, "Notes on the history of correlation," *Biometrika*, vol. 13, no. 1, pp. 25–45, 1920.
[22] J. Lee Rodgers and W. A. Nicewander, "Thirteen ways to look at the correlation coefficient," *The American Statistician*, vol. 42, no. 1, pp. 59–66, 1988.
[23] L. Labs, "Lightning network daemon (lnd)," https://github.com/lightningnetwork/lnd, 2023, accessed: 2024-11-12.
[24] V. Batagelj and U. Brandes, "Efficient generation of large random networks," *Phys. Rev. E*, vol. 71, p. 036113, Mar 2005. [Online]. Available: https://link.aps.org/doi/10.1103/PhysRevE.71.036113

Per l'implementazione usare questo modello